WORKSHEET 3.2 : CARTESIAN EQUATION OF A LINE

3a) $2x+7y+C=0$

$2(4)+7(-2)+C=0$

$8-14+C=0$

$C=6$

$\therefore \boxed{2x+7y+6=0}$

b) $-4x+0y+C=0$

$-4\left(\frac{1}{2}\right)+C=0$

$-2+C=0$

$C=2$

$\therefore -4x+2=0$

$\boxed{4x-2=0 \text{ OR } 2x-1=0}$

c) $x+y+C=0$

$3+3+C=0$

$C=-6$

$\therefore \boxed{x+y-6=0}$

d) $-x+y+C=0$

$-\frac{1}{3}+\frac{1}{3}+C=0$

$C=0$

$\therefore -x+y=0$

$\boxed{x-y=0}$

4a) $4x+3y-12=0$

$\vec{n}=(4,3)$

$\vec{m}=(3,-4)$

$\vec{n}\cdot\vec{m}=0$ ✓

Po, let $x=0$ $\therefore 3y-12=0$ $\therefore (0,4)$

$y=4$

b) $3x-6y=14$

$\vec{n}=(3,-6)$

$\vec{m}=(6,3)$

Po, let $x=1$ $3(1)-6y=14$ $\therefore \left(1,\frac{-11}{6}\right)$

$-6y=11$

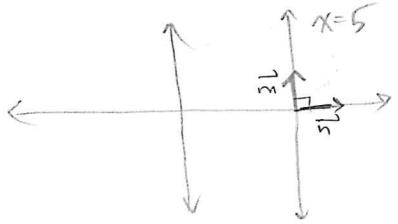
$y=\frac{-11}{6}$

c) $x = 5$

$\vec{n} = (1, 0)$

$\vec{m} = (0, 1)$

$P_0 (5, -3)$



d) $y = 3x - 10$

$0 = 3x - y - 10$

$\vec{n} = (3, -1)$

$\vec{m} = (1, 3)$

P_0, let $y = 2$

$2 = 3x - 10$

$12 = 3x$

$4 = x$

$\therefore (4, 2)$

5) If $\perp$, then dot product $= 0$

$(-b, a) \cdot (a, b)$

$= -ab + ab$

$= \emptyset$

$\therefore (-b, a) \perp (a, b)$

$(b, -a) \cdot (a, b)$

$= ab - ab$

$= \emptyset$.

$\nRightarrow$ $(b, -a) \perp (a, b)$ $\forall a, b \in \mathbb{R}$

6a) $(x, y) = (4, -6) + t(8, 2)$

$\vec{m} = (8, 2)$

$\vec{n} = (2, -8)$ or $\vec{n} = (1, -4)$

$x - 4y + C = 0$

$(4) - 4(-6) + C = 0$

$4 + 24 + C = 0$

$C = -28$

$\therefore \boxed{x - 4y - 28 = 0}$

b) $x = 3 + 18t$

$y = 4 + 9t$

$\vec{m} = (18, 9)$

$\vec{n} = (9, -18)$

or $\vec{n} = (1, -2)$

$x - 2y + C = 0$

$3 - 2(4) + C = 0$

$3 - 8 + C = 0$

$C = 5$

$\therefore \boxed{x - 2y + 5 = 0}$

6c) $\vec{r} = (2,7) + t(2,7)$

$\vec{m} = (2,7)$

$\vec{n} = (7,-2)$

$7x - 2y + C = 0$

$7(2) - 2(7) + C = 0$

$14 - 14 + C = 0$

$C = 0$

$\therefore \boxed{7x - 2y = 0}$

d) $x = 2t$, $y = -2$ $\Big\}$ $\vec{m} = (2,0)$, $\vec{n} = (0,2)$ OR $\vec{n} = (0,1)$

$\therefore y + C = 0$

$(-2) + C = 0$

$C = 2$

$\therefore \boxed{y + 2 = 0}$

7a) $\vec{n} = (2,-3)$ since $\parallel$, same $\vec{n}$

$2x - 3y + C = 0$

$2(2) - 3(-6) + C = 0$

$4 + 18 + C = 0$

$C = -22$

$\therefore \boxed{2x - 3y - 22 = 0}$

b) $\perp$ to $3x - 2y + 12 = 0$

$\vec{n_1} = (3,-2)$ $\therefore$ $\vec{n_2} = (2,3)$

$2x + 3y + C = 0$

$2(2) + 3(-6) + C = 0$

$4 - 18 + C = 0$

$C = 14$

$\therefore \boxed{2x + 3y + 14 = 0}$

7c) $\vec{m} = (2, -3)$ $\therefore \vec{n} = (3, 2)$

$3x + 2y + C = 0$

$3(2) + 2(-6) + C = 0$

$6 - 12 + C = 0$

$C = 6$

$\therefore \boxed{3x + 2y + 6 = 0}$

d) $\vec{n} = (3, -2)$

$3x - 2y + C = 0$

$3(2) - 2(-6) + C = 0$

$6 + 12 + C = 0$

$C = -18$

$\therefore \boxed{3x - 2y - 18 = 0}$

8) // to $\left.\begin{array}{l} x = -4 - 5t \\ y = 11 + 3t \end{array}\right\}$ $\vec{m} = (-5, 3)$

$\therefore$ $x = 8 - 5s$, $y = -2 + 3s$

$\frac{x-8}{-5} = s$ $\quad$ $\frac{y+2}{3} = s$

$\therefore$ symmetric Equation

$\frac{x-8}{-5} = \frac{y+2}{3}$

$3(x-8) = -5(y+2)$

$3x - 24 = -5y - 10$

$\boxed{3x + 5y - 14 = 0}$

9a) $5x - 3y + 15 = 0$

$\vec{n} = (5, -3)$

$\vec{m} = (3, 5)$

Let $x = 0$

$-3y + 15 = 0$

$y = 5$

$\therefore P_0 (0, 5)$

$\vec{r} = (0, 5) + t(3, 5)$, $t \in \mathbb{R}$

$\left.\begin{array}{l} x = 3t \\ y = 5 + 5t \end{array}\right\}$ $t \in \mathbb{R}$

$\frac{x}{3} = \frac{y-5}{5}$

b) $-4x + 6y + 9 = 0$

$\vec{n} = (-4, 6)$

$\vec{m} = (6, 4)$ OR $(3, 2)$

Let $x = 0$, $6y + 9 = 0$

$y = -\frac{3}{2}$ $\quad (0, -\frac{3}{2})$

$\vec{r} = (0, -\frac{3}{2}) + t(3, 2)$, $t \in \mathbb{R}$

$x = 3t$

$y = -\frac{3}{2} + 2t$, $t \in \mathbb{R}$

$\frac{x}{3} = \frac{y + 1.5}{2}$